

The Basic Living Economic Index (BLEI):

A Temporal Stability Framework for Post-Scarcity Economic System Evaluation

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Structured Abstract

Purpose. Standard welfare metrics — GDP per capita, Gini coefficients, and median income — quantify the stock and flow of nominal wealth but remain structurally blind to the direction of financial contracts. A household paying market rent and one accruing equivalent community housing equity appear identical in conventional accounting while experiencing diametrically opposite wealth trajectories. This paper introduces a unified temporal stability framework to correct that measurement gap, formalized as a simulation-parameter specification for agent-based modeling of alternative economic architectures.

Design / Methodology. Six complementary indices are derived: the Basic Living Economic Index (BLEI), Extractive Drain Coefficient (EDC), Effective Purchasing Power Multiplier (EPPM), Financial Bandwidth Score (FBS), Compound Stability Index (CSI), and Indirect Participant Benefit Index (IPBI). Each is mathematically formalized and applied in comparative scenario modeling against three US household profiles using 2023 Consumer Expenditure Survey and Federal Reserve data. Parameters are calibrated against deployed real-world analogues — the Alaska Permanent Fund Dividend (BU allocation), community land trusts (Acre Equity accumulation), and Mondragon Corporation (PTF cooperative overhead reduction) — and tested against the Creative Currency Octaves (CCO) and Public Trust Foundations (PTF) system architecture.

Findings. Under optimal CCO-PTF parameters, near-poverty participants achieve a 3.9× improvement in temporal stability at system entry, rising to 16.6× at twelve months and reaching the Flourishing tier — 730+ days of basic living covered, enabling multi-year entrepreneurial investment without required early-stage break-even — at approximately 33 months. Median-income households achieve Flourishing-tier stability at entry. High-wealth non-participants accrue \$22,800–\$59,900 in estimated annual indirect welfare gains through crime reduction, healthcare improvement, and property value appreciation. Sensitivity analysis confirms that BLEI scores remain positive (above Tier 2 Threshold) for CCO-PTF participants across the full tested parameter range at the 12-month mark, including ϵ as low as 1.50× and Acre Equity appreciation as low as 2%.

Originality / Value. The framework addresses three structural gaps absent from existing welfare indices: the extraction-accumulation asymmetry standard metrics treat as equivalent; the purchasing power premium of community-priced essential goods; and the temporal stability dimension governing whether households can realistically commit capital to ventures without expected break-even within the first two years. All six indices are formalized as simulation parameters for agent-based modeling. A sensitivity analysis and failure mode analysis are included to bound the conditions under which the framework's welfare improvements hold.

JEL Classification: D60 · D31 · I32 · O17 · R21 · E41 · C63

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1. Introduction

The persistence of welfare measurement failure in mainstream economic science is not primarily a data problem — it is a conceptual one. Standard welfare indicators were designed to measure the accumulation of nominal wealth within market structures that treat all financial outflows as equivalent. A dollar paid in mortgage interest and a dollar deposited into a savings account both reduce the household's cash balance by one dollar. In the language of standard accounting, they are identical. In their structural effect on household stability and long-run wealth trajectory, they are opposite.

This paper addresses three related measurement failures that become especially consequential when evaluating dual-currency and community ownership economic systems against traditional market economies.

1.1 The Extraction-Accumulation Asymmetry

Extractive financial contracts — market rent, mortgage interest, consumer debt service, and predatory fees — transfer household resources to counterparties without building household wealth. Generative instruments — equity accumulation, depreciation-adjusted ownership, cooperative share-building — redirect equivalent cash flows into household balance sheets. Standard welfare metrics assign equal negative weight to both.

In the United States, the average renter in the bottom income quintile directs approximately 62–68% of gross income to rent alone — a fully extractive transfer [9]. A comparable household participating in a Public Trust Housing (PTH) community directs a similar proportion to monthly payments, but those payments accrue entirely to the resident's Acre Equity account. Standard metrics rate both households identically.

1.2 The Basic Unit Purchasing Power Premium

In the Creative Currency Octaves (CCO) system, Basic Units (BU) are allocated monthly at approximately \$1,200 per adult and redeemable at Public Trust Foundation (PTF) establishments at below-market prices. A PTF breakfast priced at 3 Basic Units replaces a \$7.00 market-price meal; a PTF dinner at 5 BU replaces a \$15.00 market-price meal. The effective purchasing power premium of each Basic Unit averages approximately 2.64× face value at PTF food price points. This premium is grounded in the 35–55% overhead reduction achieved by cooperative institutional models such as Mondragon Corporation and documented community land trust networks (Section 3). Standard income metrics record the monthly BU allocation as equivalent to \$1,200 in income, missing the price differential that makes \$1,200 in BU worth approximately \$2,502 in effective market purchasing power.

1.3 The Temporal Stability Dimension

Maslow's hierarchy of needs carries a precise economic implication: individuals whose survival is not secured cannot allocate cognitive and financial resources toward self-actualization, creative production, or multi-year entrepreneurial investment. Mullainathan and Shafir's (2013) research on scarcity cognition demonstrates that resource deprivation taxes cognitive bandwidth directly, reducing the effective intellectual capacity available for complex planning and delayed-gratification decisions [18]. Standard welfare metrics measure the level of income or wealth at a single point in time. They do not measure how many days that wealth buys of stable basic living — the critical parameter for whether a household can realistically commit capital to a venture without requiring break-even within the first two years.

The BLEI converts wealth and income into a unified temporal stability score, expressed in days, that captures this dimension and enables the six-tier Flourishing classification introduced in Section 4.

1.4 Framework Positioning

This paper makes three distinct contributions: (1) a *measurement framework proposal* — six indices that correct structural gaps in existing welfare metrics; (2) a *simulation parameter specification* — a formalized parameter set for agent-based modeling of alternative economic architectures; and (3) a *comparative scenario analysis* — calibrated projections for three US household profiles under CCO-PTF versus baseline conditions. These contributions are not empirical studies of deployed CCO-PTF systems, which do not yet exist at scale. They are forward-looking scenario models anchored to parameters validated by real-world analogues in Section 3.

2. Existing Frameworks and Their Limitations

The inadequacy of GDP-based welfare measurement has attracted extensive scholarly attention. Sen and Anand (1994) demonstrated that income-based measures systematically misrepresent welfare where non-income dimensions diverge from economic production, motivating the Human Development Index [1,2]. The Alkire-Foster Multidimensional Poverty Index improved on binary poverty thresholds by measuring simultaneous deprivations across ten indicators [3], yet treats each dimension as a stock at a single point in time and does not model the trajectory of wealth accumulation versus extraction over time.

Buffer-stock saving theory (Carroll, 1997) established that households maintain a target level of liquid wealth as a buffer against income uncertainty, with optimal consumption behavior strongly shaped by this precautionary motive [4]. This is the closest theoretical precedent to the BLEI's temporal framing, but it does not address the distinction between extractive and generative financial contracts. Lusardi, Schneider, and Tufano (2011) established that a majority of US households could not absorb a \$2,000 financial shock without borrowing [5] — precisely the condition the BLEI formalizes as the Precarious tier.

Mullainathan and Shafir (2013) demonstrated experimentally that scarcity — defined as the subjective experience of having less than one needs — directly taxes cognitive bandwidth and

impairs decision-making quality in ways that compound poverty traps [18]. This behavioral mechanism provides the psychological foundation for the BLEI's Flourishing threshold: below a certain temporal security level, households cannot realistically commit to multi-year investments regardless of nominal income.

Piketty's (2014) analysis of wealth concentration dynamics established that when the return on capital (r) exceeds economic growth (g), wealth inequality compounds over time [19]. The BLEI framework extends this insight at the household level: under extractive contracts, the household's effective r is negative (rent payment destroys wealth); under generative contracts (PTH Acre Equity), r is positive (monthly payments build compounding equity). The divergence between these trajectories is the core measurement gap the EDC and CSI are designed to capture.

Financial health measurement frameworks — including the CFPB Financial Well-Being Scale [20] and the US Financial Health Pulse survey — identify similar dimensions of resilience, day-to-day management, and forward-looking capacity. The BLEI differs from these primarily in its explicit modeling of the generative/extractive contract distinction, the dual-currency purchasing power adjustment, and its formalization as an agent-based simulation parameter rather than a survey instrument.

None of these frameworks addresses the purchasing power premium generated by community-priced essential goods, nor do they model the compounding divergence between generative and extractive financial contracts at multi-year horizons. The BLEI framework is designed to address all three gaps.

3. Real-World Analogues and Parameter Grounding

The CCO-PTF system as a whole does not yet exist at national scale, but its three core mechanisms each have direct real-world analogues that provide empirical grounding for key parameters. This section documents those analogues and their implications for the BLEI framework.

3.1 Alaska Permanent Fund Dividend — BU Allocation Analogue

Deployed System: Alaska Permanent Fund (1982–present)

The Alaska Permanent Fund Dividend (APF) distributes oil revenue to all Alaska residents unconditionally: \$1,312–\$2,072 per year (\$109–\$173 per month) in recent years. Jones and Marinescu (2018) found that APF increased part-time work by 1.8 percentage points with no reduction in full-time employment — directly contradicting dependency-effect predictions [21]. Over 40 years of operation, the APF has shown no significant negative labor supply effects despite covering a meaningful share of basic living costs.

Parameter implication: BU allocation of \$1,200 per month is approximately $7\times$ the APF annual dividend. The CCO system operates at fundamentally different scale, requiring the PTF ecosystem as the price-control mechanism to maintain $\varepsilon \geq 2.0\times$. The APF data validates the

absence of dependency effects from universal distributions; the PTF cooperative pricing mechanism is the CCO-specific innovation enabling the purchasing power premium.

3.2 Community Land Trusts — Acre Equity Analogue

Deployed System: ~300 CLTs across 46 US states (National CLT Network, 2023)

Community Land Trusts separate land ownership (held by the trust in perpetuity) from home ownership (held by residents), enabling below-market housing with equity accumulation. The Burlington Community Land Trust (founded 1984) enables residents to pay 25–30% of income for housing versus 40–50% at market rate while building equity through a shared appreciation formula. Theodos et al. (2017) found CLT homeowners have 70% lower foreclosure rates than conventional mortgages [22]. Davis and Demetrowitz (2003) documented average CLT resident equity at resale of \$13,000–\$22,000 after ten years [23].

Parameter implication: The Acre Equity accumulation mechanism is functionally equivalent to CLT shared equity, with the PTH community replacing the land trust as the institutional owner. The CLT literature validates: (a) the feasibility of below-market housing payments that simultaneously build resident equity, (b) r_a values of 2–5% as consistent with long-run CLT appreciation data, and (c) the governance stability of democratically managed housing communities.

3.3 Mondragon Corporation — PTF Cooperative Economics Analogue

Deployed System: Mondragon Corporation, Basque Country (1956–present)

Mondragon comprises 80+ worker-owned cooperatives with approximately \$12B annual revenue and 80,000+ worker-owners. Errasti et al. (2003) and Whyte and Whyte (1991) document overhead cost reductions of 35–55% relative to comparable conventional corporations, with operating margins retained within the cooperative rather than extracted by external shareholders [24,25]. Mondragon's food cooperative (Eroski) operates at gross margins 30–45% below conventional supermarket chains through collective purchasing and non-profit distribution logic.

Parameter implication: The PTF's 40–60% overhead reduction assumption — which generates the BU purchasing power premium $\varepsilon \geq 2.0\times$ — is grounded in Mondragon's operational data. The $\varepsilon = 2.64$ base case represents a conservative estimate; Mondragon's food operations suggest ε values of 2.5–3.0 $\times$ are achievable at mature cooperative scale. Early-phase PTF operations may achieve ε closer to 1.5–2.0 $\times$ before achieving Mondragon-equivalent efficiency.

4. The BLEI Index Suite: Mathematical Formalization

4.0 Stock-Flow Accounting and Measurement Scope

The BLEI is a *liquidity horizon* measure: the number of days of basic living that a household's accessible economic resources can cover given current income flows and asset stocks. It is a

hybrid measure by design, combining (a) liquid asset stock L — what the household holds — with (b) a liquidity-adjusted income term $\gamma \cdot Y$ — what the household can access as a buffer from ongoing income — and (c) a BU effective value term where applicable. The Acre Equity component carries a liquidity haircut reflecting the settlement lags and exit mechanics of PTH community equity (see Table 1a). This hybrid construction reflects how households actually experience financial resilience: both "how much do I have" and "how long can my income sustain me" are relevant to the temporal stability question.

Index I — Basic Living Economic Index (BLEI)

The BLEI expresses the number of days of basic living covered by a household's accessible economic resources, accounting for liquid assets, liquidity-adjusted income, and Basic Unit allocation where applicable.

Core formula:

$$\text{BLEI}(i,t) = [L(i,t) + \gamma \cdot Y(i,t) + \text{BU}(i,t) \cdot \varepsilon(t)] / C_{\text{basic}}(t)$$

$L(i,t)$ = liquid assets of household i at time t , including AE at liquidity haircut h_{AE} :

$$L = \text{cash_savings} + h_{\text{AE}} \cdot \text{AE}(t)$$

where h_{AE} varies by PTH tenure (see Table 1a)

γ = liquidity coefficient: fraction of monthly income accessible as buffer
 0.12 baseline (no guaranteed security floor; income consumed by extraction)
 0.20 under CCO (guaranteed BU floor reduces income-vulnerability; higher γ supported by Carroll (1997) buffer-stock theory: guaranteed floors expand the accessible income fraction)

$Y(i,t)$ = monthly primary-currency income

$\text{BU}(i,t)$ = monthly Basic Unit allocation (0 in baseline; \$1,200/adult under CCO)
 FLAT allocation — NOT scaled by octave level.
 Octave controls BU→primary currency conversion capacity, not allocation amount.

$\varepsilon(t)$ = BU effective purchasing power coefficient at PTF prices
 Food basket (360 BU/month): $\varepsilon_{\text{food}} = 2.64$ [base case; range 1.50–3.00]
 Utilities/services (840 BU): $\varepsilon_{\text{util}} = 1.80$ [base case; range 1.40–2.50]
 Monthly effective value per adult: $360 \times \$2.75 + 840 \times \$1.80 = \$990 + \$1,512 = \$2,502$

C_{basic} = daily cost of basic living in primary currency
 Baseline: $(\$1,300 \text{ rent} + \$400 \text{ food} + \$150 \text{ utilities} + \$200 \text{ transport}) / 30 = \$68.33/\text{day}$
 CCO-PTF: $(\$600 \text{ PTH} + \$150 \text{ utilities} + \$200 \text{ transport}) / 30 = \$31.67/\text{day}$
 (food covered by BU allocation; excluded from cash denominator under CCO)

Result: BLEI expressed in days of basic living covered. Higher = greater temporal stability.

Month 0–6	0.10–0.20	10–20% of face value	Settlement lags; exit fees; minimal secondary market
Month 7–24	0.30–0.50	30–50% of face value	Partial liquidity; exit fee reduction; early market pricing
Month 25–60	0.60–0.75	60–75% of face value	Established equity; clear market value; standard exit options
Month 60+	0.80–0.90	80–90% of face value	Mature equity; well-defined secondary market; reduced exit risk

Stylized profiles in Section 7 use $h_{AE} = 0.60$ for medium-term projections (consistent with Month 25–60 cohort). Sensitivity tests across $h_{AE} \in [0, 0.90]$ are reported in Section 5.

BLEI Temporal Stability Tiers

Tier
BLEI Range
Characterization
0 — Crisis
< 7 days
Immediate vulnerability; survival-mode cognition (Mullainathan & Shafir, 2013); zero advancement capacity
1 — Precarious
7–30 days
One disruption from crisis; zero financial bandwidth (Lusardi et al., 2011 \$2,000-shock threshold); income fully consumed by extraction and basic needs
2 — Threshold
30–120 days
BLEI-defined poverty line; limited Maslow advancement feasible; positive FBS emerging

3 — Stable

120–365 days

Sufficient buffer for measured risk-taking and education investment; first year covered

4 — Secure

365–730 days

Resilient; creative and entrepreneurial participation feasible within a two-year payoff horizon

5 — Flourishing

> 730 days

Two-year capital commitment horizon unlocked; ventures without expected payoff within two years are viable; extended planning horizon with reduced liquidity-constrained decision-making

The Flourishing tier threshold of 730 days — two full years of basic living covered — reflects the capital commitment horizon required for entrepreneurial ventures to become viable without requiring break-even within the first year. This is the threshold at which multi-year creative and entrepreneurial investment becomes structurally possible, not merely aspirationally imaginable.

PTH Acre Equity accumulation. Under Public Trust Housing, monthly payments (H) accrue entirely to the resident's Acre Equity balance. After n months:

$$AE(n) = H \cdot [(1 + r_a/12)^n - 1] / (r_a/12)$$

H = monthly PTH payment (\$600 at base case parameters)

r_a = annual Acre Equity appreciation rate

Base case: 0.04 (conservative; consistent with CLT long-run data)

Sensitivity range: 0.02–0.07

n = months of participation

AE enters L(i,t) at liquidity haircut h_AE (see Table 1a).

Under market rent, H is fully extractive and does not enter L(i,t) at any horizon —

the fundamental asymmetry standard metrics cannot see.

Index II — Extractive Drain Coefficient (EDC)

The EDC measures the proportion of gross monthly income captured by financial contracts that generate no household wealth accumulation. It is a diagnostic index: high EDC indicates that a large share of income is being permanently transferred to counterparties with no balance-sheet return to the household.

$$EDC(i) = \frac{[\text{Rent_full} + \text{Mortgage_Interest}(i) + \text{Consumer_Debt_Interest}(i) + \text{Fees}(i)]}{Y(i) \quad (\text{monthly gross income})}$$

Component definitions:

Rent_full: full rental payment (100% extractive; zero equity component)

Mortgage_Interest: interest-only portion of mortgage payments; principal excluded

(principal payments build equity and are therefore NOT included)

Consumer_Debt_Interest: credit card, auto loan, personal loan, student loan interest

Fees: payday loan fees, overdraft charges, predatory service costs

Scope note: property taxes, insurance, maintenance, healthcare premiums, childcare, and income taxes are structural costs that affect disposable income but are not classified as "extractive" under this framework because they fund collective goods or social insurance, not private wealth transfers. They are incorporated in C_basic where applicable.

EDC target under CCO-PTF: ~0.025 (PTH payments generative; BU reduces debt reliance)

US baseline near-poverty: ~0.625-0.72

US baseline median: ~0.32-0.40

EDC Liberation Differential (annual): $\Delta EDC = EDC_baseline \times Y(i) \times 12$

Near-poverty (\$2,083/mo, EDC = 0.625): \$15,623/year previously extracted

Index III — Effective Purchasing Power Multiplier (EPPM)

The EPPM is a dimensionless ratio expressing the effective purchasing power of CCO-PTF household income relative to a baseline household at the same nominal income level, after accounting for BU allocation, EDC reduction, and PTF price differential. EPPM = 1.00 in the absence of CCO-PTF.

$$EPPM(i) = \frac{Y(i) + BU(i) \cdot \varepsilon_blended - EDC_CCO \cdot Y(i)}{Y(i) \cdot (1 - EDC_baseline)}$$

$BU(i) \cdot \varepsilon_blended$ = total effective market purchasing power of monthly BU allocation
 = (360 BU food × \$2.75 market equiv) + (840 BU utilities × \$1.80 PTF equiv)
 = \$990 + \$1,512 = \$2,502 per adult per month

Numerator = net disposable resources under CCO-PTF (income + BU value - residual extraction)

Denominator = net disposable resources under baseline (income after extractive drain)

Unit: dimensionless ratio (EPPM > 1.0 indicates improvement; < 1.0 would indicate degradation)

At poverty line ($Y = \$2,083$, $EDC_CCO = 0.025$, $EDC_baseline = 0.625$):

$$EPPM = (\$2,083 + \$2,502 - \$52) / (\$2,083 \times 0.375) = \$4,533 / \$781 = 5.80\times$$

Index IV — Financial Bandwidth Score (FBS)

The FBS measures monthly residual resources available above the basic living threshold. Drawing on Mullainathan and Shafir's (2013) cognitive bandwidth research, positive FBS is the mechanism through which temporal security translates into advancement capacity: without slack, planning bandwidth is consumed by immediate survival [18]. Without positive FBS,

investment in education, creative work, or business formation is structurally infeasible regardless of individual capability.

Component clarification (no double-counting). The `C_basic_cash_monthly` term in FBS is defined to exclude any component already captured in EDC. Under CCO-PTF, housing cost is included in `C_basic` as PTH (\$600); therefore EDC in the FBS calculation represents *residual consumer debt interest only* — not housing. Under the baseline, where rent is included in `C_basic`, EDC in the FBS calculation is similarly restricted to consumer debt interest. This avoids the double-subtraction of extractive flows.

Y (income)	\$2,083/mo	\$2,083/mo
BU (monthly allocation)	\$0	\$1,200/mo
C_basic_cash_monthly	\$1,950 (rent \$1,300 + food \$400 + util \$150 + transport \$200; rounded)	\$950 (PTH \$600 + util \$150 + transport \$200; food covered by BU)
EDC_residual × Y (FBS context)	Consumer debt interest only: ~\$250/mo (EDC_residual ≈ 0.12 × \$2,083)	Consumer debt interest only: ~\$52/mo (EDC_residual ≈ 0.025 × \$2,083)
FBS = max(0, Y + BU – C_basic – EDC_res · Y)	max(0, \$2,083 – \$1,950 – \$250) = \$0 (deficit: –\$117)	max(0, \$2,083 + \$1,200 – \$950 – \$52) = \$2,281/mo
$\text{FBS}(i,t) = \max(0, Y(i) + \text{BU}(i) - \text{C_basic_cash_monthly}(t) - \text{EDC_residual}(i) \cdot Y(i))$		

where `EDC_residual` = consumer debt interest only; housing costs are in `C_basic_cash_monthly` and must not appear in both terms simultaneously.

Advancement probability: $P(\text{advance}) = 1 - \exp(-\lambda \cdot \text{FBS})$
where λ = individual capability coefficient ~ U(0.001, 0.008)
Agents with `FBS` = 0 have $P(\text{advance}) \approx 0$ regardless of quality score.

This is the behavioral mechanism documented in scarcity cognition research [18].

Index V — Compound Stability Index (CSI)

The CSI projects the intertemporal trajectory of household stability by compounding initial BLEI by net wealth growth rate, adjusted for extractive drain, and adding accumulated Acre Equity normalized to daily costs. It is intended as a forward-looking stability horizon, not a stock measure at a single point in time.

$$\text{CSI}(i,t) = \text{BLEI}(i,0) \cdot (1 + r_{\text{wealth}})^t \cdot (1 - \text{EDC}(i)) + \text{AE}(i,t) / \text{C_basic}(t)$$

All terms measured in days of basic living coverage (consistent unit throughout):

$\text{BLEI}(i,0) \cdot (1 + r_{\text{wealth}})^t \cdot (1 - \text{EDC})$:
= initial stability horizon, compounded by net wealth growth rate,
reduced by extractive drain fraction

$\text{AE}(i,t) / \text{C_basic}(t)$:
= Acre Equity accumulated through month t , expressed as days of coverage
at current daily cost (both terms in USD; ratio is dimensionless days)

r_{wealth} under baseline renter: ~0.01-0.03 (savings growth only; no equity)
 r_{wealth} under CCO-PTH: ~0.08-0.15 (Acre Equity + BU buffer investment)

Derivation note: CSI is a simplified intertemporal projection grounded in the identity $\text{W_accessible}(t) = \text{W_accessible}(0) \cdot (1 + r_{\text{eff}})^t$, where $r_{\text{eff}} = r_{\text{wealth}} \cdot (1 - \text{EDC})$ and AE is tracked separately as a generative-only flow not captured in W_accessible . The two-term structure reflects the separation between liquid wealth trajectory and

illiquid (but appreciating) equity trajectory – both in days-of-coverage units.

Index VI — Indirect Participant Benefit Index (IPBI)

The IPBI quantifies estimated annual welfare gains accruing to high-wealth households who do not directly participate in CCO-PTF but benefit from systemic improvements generated by broad poverty elimination. IPBI channels are treated as additive for estimation purposes, but may exhibit partial overlap (notably crime reduction and property value channels); researchers conducting causal inference studies should partition channels cleanly and test for double-counting within their specific context. The estimates below are order-of-magnitude ranges derived from the macroeconomic cost-of-poverty literature and should not be treated as precise household-level projections.

$$\text{IPBI} = \alpha \cdot \text{CR} + \beta \cdot \text{PH} + \gamma \cdot \text{LP} + \delta \cdot \text{PV} + \zeta \cdot \text{ME}$$

CR = crime-reduction benefit (security savings + insurance reduction)
Anderson (1999) estimated total US crime costs at ~\$1.0T/year [6];
30-50% poverty-correlated share; 35-55% reduction estimate per household:
\$3,500-\$9,000/year

PH = public health benefit: healthcare cost reduction from poverty relief
Capretta (2022) documents poverty-linked healthcare premium [15]:
\$800-\$2,400/year per HH

LP = labor productivity benefit: 11-14% gain from workforce financial stress elimination
PwC Employee Financial Wellness Survey (2023) [8]: \$4,000-\$12,000/year per employer HH

PV = property value appreciation: 2-5% on median \$650K high-wealth home
Greenbaum & Tita (2004) on crime-reduction neighborhood effects [7]:
\$13,000-\$32,500

ME = market expansion: positive-FBS consumer base growth: \$1,500-\$4,000/year

Note on potential channel overlap: PV appreciation may partially reflect CR effects (crime reduction raises property values). In the high-IPBI estimate (\$59,900), treating CR and PV as independent likely overstates by an estimated 10-20%. Conservative estimate (\$22,800) treats PV as the residual channel and is more defensible.

Total estimated IPBI: \$22,800-\$59,900/year (zero direct participation required)

5. Sensitivity Analysis

The following table reports the sensitivity of BLEI outcomes (Profile A: near-poverty single adult) to variation in key parameters across their tested ranges. The analysis reveals that ϵ is the dominant parameter at short horizons (Month 0-6); r_a dominates at longer horizons (Month 36+). The framework produces BLEI scores above the Tier 2 Threshold (30 days) for CCO-PTF participants across all tested parameter combinations at Month 12 — the result that is most relevant for policy evaluation.

ϵ (BU purchasing power)	2.64×	1.50×	3.00×	71.3 → 47.2 → 103.4 days	303 → 213 → 392 days
r_a (AE appreciation)	4% / yr	2% / yr	7% / yr	71.3 → 68.1 → 74.6 days	303 → 289 → 322 days
γ (liquidity coefficient)	0.20	0.12	0.28	71.3 → 60.0 → 82.7 days	303 → 264 → 342 days
h_{AE} (AE liquidity haircut)	0.60	0.00	0.90	71.3 → 65.9 → 74.5 days	303 → 233 → 341 days

C_basic daily (CCO-PTF)	\$31.67/d ay	\$28.00/d ay	\$45.00/day	71.3 → 80.6 → 50.2 days	303 → 343 → 213 days
C_basic regional (high-cost metro)	\$31.67/d ay	—	\$52.00/day (SF/NYC)	71.3 days	303 → — → 130 days (Stable)

Table 2. BLEI sensitivity analysis — Profile A (near-poverty, \$2,083/month). All combinations maintain BLEI $\geq$ Tier 2 Threshold (30 days) at Month 12 except the highest-cost urban region at lowest ϵ ($\epsilon=1.50$, $C_basic=\$52/\text{day}$ → BLEI Mo. 12 $\approx$ 49 days; still above threshold but in lower range).

Key finding: The most vulnerable scenario combines low ϵ ($1.50\times$, early-phase PTF), high regional cost (\$45–52/day), and low AE haircut realization (0–20%). Even under these conditions, CCO-PTF participants maintain BLEI above 35 days at Month 12 — above the Tier 2 Threshold — due to the combined effect of reduced housing cost (\$600 vs. \$1,300) and the positive FBS generated by BU allocation even at reduced ϵ . The Flourishing tier (730 days) is reached between 40 and 55 months across the full sensitivity range for Profile A.

6. System Failure Modes and Risk Considerations

A welfare measurement framework that only performs under optimal conditions provides limited value for policy design. The following failure modes are identified as conditions under which the BLEI framework's projected improvements would be substantially degraded. Mitigation strategies are described; full treatment appears in the companion Risk Mitigation Framework paper [16].

6.1 Insufficient PTF Network Density (ϵ Degradation)

Below approximately 40% PTF merchant participation in a given zone, the BU purchasing power premium ϵ falls toward $1.5\times$ – $1.8\times$ as insufficient volume prevents Mondragon-equivalent overhead reduction. The simulation network-density-gated synergy coefficient θ addresses this: below 55% participation density, cooperative synergies do not activate ($\theta = 0$). *Mitigation:* Phase deployment beginning in high-density urban zones; SZH zone activation thresholds; PTF investment subsidy during early scaling. BLEI outcomes at $\epsilon = 1.50$ remain above Threshold tier (Table 2).

6.2 Governance Capture

PTF and PTH democratic governance can be captured by organized factions that redirect community resources toward narrow interests, degrading service quality and AE value. *Mitigation:* CIP transparent blockchain voting; supermajority requirements for major expenditure decisions; cross-zone audit oversight; interpersonal financial abuse protections for residents (documented in Risk Mitigation Framework [16]).

6.3 Participation Below Critical Mass

Below approximately 60% CCO participation rate, conversion tax revenue may be insufficient to fund the \$1,200/adult BU allocation at national scale without external fiscal support.

Mitigation: Phased deployment in targeted communities; parallel government subsidy during transition; voluntary participation incentives reducing the effective participation threshold required for self-funding.

6.4 Inflationary Pressure in Non-PTF Markets

BU-stimulated demand without PTF price anchoring could raise prices in non-essential markets, potentially increasing C_basic for goods outside the PTF system. The dual-currency design (BU restricted to essentials; 12% conversion tax on CCO-to-fiat) limits this channel.

Mitigation: PTF market share target of 18% for essential goods provides sufficient price anchoring for the BLEI C_basic denominator; non-essential market inflation does not affect BLEI unless households consume non-essentials with primary currency resources that would otherwise fund C_basic.

6.5 Acre Equity Liquidity Risk

If PTH communities experience governance failure, physical deterioration, or housing market decline, AE values may not be realizable at projected levels. The liquidity haircut table (Table 1a) partially addresses this by discounting AE in early tenure periods. *Mitigation:* Maintenance reserve requirements; insurance backstop mechanisms; CIP democratic oversight with capital reserve requirements (see Risk Mitigation Framework [16]). Sensitivity analysis (Table 2) shows that h_AE = 0.00 (complete illiquidity) still produces Tier 3–4 BLEI outcomes at Month 12 due to the FBS channel.

7. Comparative Scenario Analysis: CCO-PTF vs. Current US Economy

We apply the BLEI suite to three US household profiles using 2023 Consumer Expenditure Survey data as the baseline [9,10,11]. These are *calibrated scenario models*, not observations of existing CCO-PTF households. They demonstrate the framework's measurement properties and the implied welfare gap between systems under optimal CCO-PTF parameters — the conditions identified in the companion simulation replication framework [13] as achievable with full deployment. CCO-PTF parameters: \$1,200/adult/month BU; PTF meal pricing 3/4/5 BU (breakfast/lunch/dinner); PTH monthly payment \$600 (100% generative, r_a = 0.04). All figures in 2023 USD.

7.1 Profile A — Near-Poverty Single Adult

Monthly income: \$2,083 (25th wage percentile). Renter. Liquid savings: \$1,000. AE haircut h_AE = 0.60 applied to medium-term projections.

Monthly income	\$2,083	\$2,083	—
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BU allocation (effective market value)	\$0	\$2,502/mo (BU × $\varepsilon_{\text{blended}}$)	+\$2,502
Housing / type	\$1,300 rent (extractive)	\$600 PTH (builds Acre Equity)	−\$700 cash; \$600 building AE
Monthly food cost	\$400 cash	360 BU from allocation	\$400 freed from cash
EDC	0.625	0.025	−60 points
Daily cash basic cost	\$68.33/day	\$31.67/day	−54%
BLEI — Month 0	18.3 days (Tier 1: Precarious)	71.3 days (Tier 2: Threshold)	3.9×
BLEI — Month 12	18.3 days (Tier 1: Precarious)	303 days (Tier 3: Stable)	16.6×
BLEI — Month 33	18.3 days (no change)	~730 days (Tier 5: Flourishing entry)	Tier 5 at ~2.8 years
BLEI — Month 60	18.3 days (no change)	~1,328 days (Tier 5: Flourishing)	72.6×
Financial Bandwidth Score (FBS)	\$0/mo (deficit)	\$2,281/mo	+\$2,281 advancement capital
EPPM (dimensionless ratio)	1.00×	5.80×	+480%

Table 3. BLEI metrics — Near-poverty profile, CCO-PTF vs. baseline (2023 USD). Optimal CCO-PTF parameters; $h_{\text{AE}} = 0.60$ for projections beyond Month 6.

The Acre Equity compound effect. Under the baseline, BLEI is structurally static at 18.3 days: rent extracts \$1,300/month, the \$0 FBS leaves no savings buffer, and there is no accumulation mechanism. Under

CCO-PTF, \$600/month accrues to Acre Equity compounding at 4% annually. By Month 33 (approximately 2.8 years), accumulated equity combined with ongoing FBS brings BLEI above 730 days, entering the Flourishing tier. The ratio is not constant — it is exponential. The baseline household does not approach this tier through any endogenous mechanism.

7.2 Profile B — Median US Household (Two Adults, Family of Four)

Monthly income: \$6,215 (\$74,580/year, Census 2023). Homeowner, 30-year mortgage at 7% (\$420K home). Year-1 interest: \$1,960/month. Liquid savings: \$30,000.

Monthly income	\$6,215	\$6,215	—
BU allocation (2 adults, effective)	\$0	\$5,004/mo effective	+\$5,004
Housing cost / type	\$2,236 P&I (\$1,960 interest extractive)	\$600/mo PTH, 100% generative	\$1,960 extraction eliminated
Monthly food cost	\$717 cash	720 BU from allocation	\$717 freed from cash
EDC	0.364	0.030	−33 points
Daily cash basic cost	\$104.03/day	\$43.40/day	−58%
BLEI — Month 0	296 days (Tier 3: Stable)	759 days (Tier 5: Flourishing)	2.57× — Flourishing at entry
BLEI — Month 60	~392 days (Tier 4: Secure)	~1,675 days (Tier 5: Flourishing)	4.27×
EPPM	1.00×	2.79×	+179%

Table 4. BLEI metrics — Median household profile, CCO-PTF vs. baseline (2023 USD).

Median household Flourishing at entry. Three simultaneous effects converge: \$5,004/month in effective BU purchasing power covers all food and significant utilities; eliminating \$1,960/month in mortgage interest restores the largest single extractive outflow; and the daily cash cost halves from \$104 to \$43, extending every dollar of liquid savings 2.4× further. The FBS gain of \$3,400+/month enables children's education, business investment, and creative participation that the mortgage extraction rate structurally prevents. This is the condition — Flourishing at entry for median households — that the framework identifies as the optimal design target.

7.3 Profile C — High-Wealth Non-Participant (IPBI)

Annual income: \$200,000. No direct CCO-PTF participation. Estimated welfare gains through systemic improvements generated by broad poverty reduction. Channel estimates are conservative figures from the macroeconomic literature; see Section 4, Index VI, for channel overlap caveats.

Crime reduction (security savings + insurance)	\$3,500–\$9,000	35–55% crime reduction × household security expenditure [6]
Health insurance premium reduction	\$800–\$2,400	System-wide cost relief from poverty-linked healthcare reduction [15]
Property value appreciation	\$13,000–\$32,500	2–5% appreciation on \$650K median high-wealth home [7]
Labor productivity gain (employer context)	\$4,000–\$12,000	11–14% workforce productivity gain from financial stress elimination [8]
Market expansion (positive-FBS consumer base)	\$1,500–\$4,000	Enlarged consumer base with advancement capital
Total annual IPBI	\$22,800–\$59,900	Zero direct participation required; conservative lower bound preferred

Table 5. IPBI estimates — High-wealth non-participating household. Conservative lower bound (\$22,800) is more defensible; upper bound (\$59,900) treats all channels as independent and likely overstates by 10–20%.

8. Simulation Parameter Implications

The BLEI suite generates a revised parameter set for agent-based modeling of the CCO-PTF system. Four corrections are required in any simulation that previously used standard welfare metrics.

First, PTH payments must route to an Acre Equity state variable rather than being subtracted from liquid wealth. The symmetric-extraction error — modeling PTH and market rent as equivalent outflows — overstates baseline welfare and renders the two systems economically indistinguishable in long-run projections. Second, BU allocation should be modeled at effective purchasing power ($BU \times \varepsilon_{\text{blended}} = \$2,502/\text{adult}/\text{month}$) rather than face value (\$1,200), as the PTF price differential constitutes a real welfare gain. Third, BU allocation must be set as a flat parameter independent of octave level: octave controls conversion capacity, not basic unit allocation. Fourth, agent advancement probability must be gated by FBS: $P(\text{advance}) = 1 - \exp(-\lambda \times \text{FBS})$, so that agents with zero financial bandwidth cannot advance regardless of quality score.

Poverty definition	Income > poverty line	BLEI $\geq$ 30 days (Tier 2: Threshold)
Stability target	Not specified	BLEI $\geq$ 365 days (Tier 4: Secure)
Flourishing target	Not specified	BLEI $\geq$ 730 days (Tier 5: Flourishing)
BU value in model	\$1,200 face value	\$2,502 effective market value ($BU \times \varepsilon_{\text{blended}}$)
BU scaling with octave	$BU \times \text{octaveCap}$ (error)	Flat \$1,200/adult; octave controls conversion capacity only
PTH payment routing	Subtract from liquid wealth	Add to Acre Equity at liquidity haircut h_{AE} (Table 1a)
Gini metric	Nominal wealth	EDC-adjusted net wealth: $W_{\text{net}} = W_{\text{nominal}} - (\text{EDC} \times Y \times 12)$

Advancement gating	Quality score only	FBS-gated: $P(\text{advance}) = 1 - \exp(-\lambda \times \text{FBS})$
Synergy coefficient θ	Uniform 0.15–0.25	Network-density-gated: 0 below 55% PTF density; scales to 0.25 at 90%

Table 6. Revised simulation parameters derived from BLEI framework.

9. Discussion

The BLEI framework suggests a reframing of the central question in welfare economics. Rather than asking how much people earn or how equal the distribution is — both static and direction-blind — the relevant questions become: how many days of stable basic living does this household's total economic position support? And is that number growing or shrinking over time?

The near-poverty household earning \$2,083/month under the baseline and the near-poverty household earning \$2,083/month under optimal CCO-PTF conditions look identical on a paycheck. They experience radically different economic realities: the first operates at 18.3 days of stability with zero financial bandwidth; the second at 71.3 days and rising, with \$2,281/month of advancement capital available for the first time in that household's history. The difference is not income — it is the direction and destination of financial flows.

The Flourishing tier threshold of 730 days is particularly important for policy design. It represents the point at which a household can commit capital to a venture without requiring break-even within two years — the standard viability horizon for most small business and creative enterprise. Under the baseline, the near-poverty household never approaches this tier through any endogenous mechanism. Under CCO-PTF, they reach it within 33 months through the compound divergence between Acre Equity accumulation and extractive rental stagnation — without any increase in primary-currency income.

The IPBI analysis complicates the standard political economy framing of redistributive systems as zero-sum transfers. High-wealth households benefit by \$22,800–\$59,900 annually from systemic poverty elimination through indirect channels. The claim that CCO-PTF implementation represents a pure wealth transfer from rich to poor requires significant qualification: it may more accurately represent a correction of a negative-externality market failure, in which poverty generates social costs borne diffusely by all market participants but not priced into the financial contracts that sustain poverty.

The comparison to Nordic welfare states is instructive. The Nordic post-redistribution Gini range of 0.27–0.29 represents the best sustained outcome under existing institutional architectures. The CCO-PTF simulation's EDC-adjusted Gini target of 0.22 falls meaningfully below this range — but it is achieved through a structurally different mechanism: not redistribution of after-tax income, but transformation of the financial contracts that govern the

direction of economic flows before redistribution. The Alaska Permanent Fund demonstrates that universal distributions are institutionally viable; the CLT literature demonstrates that community equity accumulation is financially sustainable; Mondragon demonstrates that cooperative overhead reduction of the magnitude required by $\varepsilon \geq 2.0\times$ is operationally achievable. The CCO-PTF system proposes integrating these three proven mechanisms at national scale.

9.1 Limitations and Future Work

The ε coefficient is derived from representative PTF pricing targets calibrated against Mondragon food cooperative data. Early-phase PTF establishments with smaller scale may achieve ε values closer to $1.5\text{--}2.0\times$ rather than 2.64 . Sensitivity analysis (Table 2) confirms that BLEI outcomes remain above the Threshold tier even at $\varepsilon = 1.50$. The population-level distribution of BLEI outcomes across the full US income distribution — beyond the three stylized profiles presented here — represents an important extension; the companion simulation replication framework [13] provides Monte Carlo results across 10,000 parameter draws that partially address this gap. General equilibrium effects — inflation in non-PTF sectors, housing market adjustment, labor supply responses to BU allocation — are modeled at the first-order level through sensitivity analysis but are not subject to full structural estimation; this remains the most significant limitation for policy forecasting applications. No causal identification strategy is claimed for the IPBI estimates, which are order-of-magnitude derivations from macroeconomic literature rather than causal estimates from CCO-PTF deployment data. As pilot programs develop, quasi-experimental methods (difference-in-differences, synthetic control) should be applied to refine these parameters.

10. Conclusion

This paper introduced six complementary indices — BLEI, EDC, EPPM, FBS, CSI, and IPBI — constituting a unified temporal stability framework for evaluating post-scarcity economic systems against conventional market economies. The framework corrects three structural gaps absent from existing welfare measurement: the extraction-accumulation asymmetry; the purchasing power premium of community-priced essential goods; and the temporal stability dimension governing Maslow-tier advancement feasibility.

Applied to optimal CCO-PTF conditions against current US trajectories, the framework reveals a welfare gap between systems substantially larger than Gini-based comparisons suggest, growing exponentially over time due to compound divergence between generative equity accumulation and extractive contract stagnation. Near-poverty households reach the Flourishing tier — enabling genuine multi-year entrepreneurial and creative commitment — within approximately 33 months under optimal conditions. Median households achieve it immediately. High-wealth non-participants benefit by an estimated \$22,800–\$59,900 annually without direct participation.

Key parameters are grounded in three deployed real-world analogues: the Alaska Permanent Fund Dividend validates universal BU allocation without dependency effects; community land trust data validates the Acre Equity accumulation mechanism and r_a calibration; and

Mondragon Corporation validates the cooperative overhead reduction underlying $\varepsilon \geq 2.0\times$. Sensitivity analysis confirms that BLEI improvements persist across the full tested parameter range. A failure mode analysis identifies the conditions — low network density, governance capture, participation below critical mass — under which system performance would be degraded, with mitigation strategies documented in the companion Risk Mitigation Framework [16].

The six indices are formalized as simulation parameters addressing the symmetric-extraction error, BU face-value misvaluation, octave-scaling error, and FBS-advancement decoupling in agent-based models. Together they constitute a measurement toolkit for evaluating the welfare properties of alternative economic architectures — tools that the current welfare measurement literature lacks, and whose absence makes the cost of the status quo systematically invisible.

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Author's Note on Normative Position

This paper makes its normative claims openly. Economic suffering is real and, where structurally avoidable, constitutes a system failure at best—and engineered predation at worst—rather than a natural condition. The BLEI framework was developed to identify the optimal conditions under which current failures can be corrected and economic exploitation eradicated. The quantitative scenarios presented are agent-based simulation outcomes calibrated against deployed real-world analogues, not predictions of outcomes in systems not yet built. The normative goal animating this research is explicit: to secure temporal basic economic freedom for all households, ensuring that economic stability is a foundational right rather than a privilege reserved for the fortunate, the industrious, or those willing to exploit others. Independent researchers are free to state this plainly, and here, that liberty is taken to heart.

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